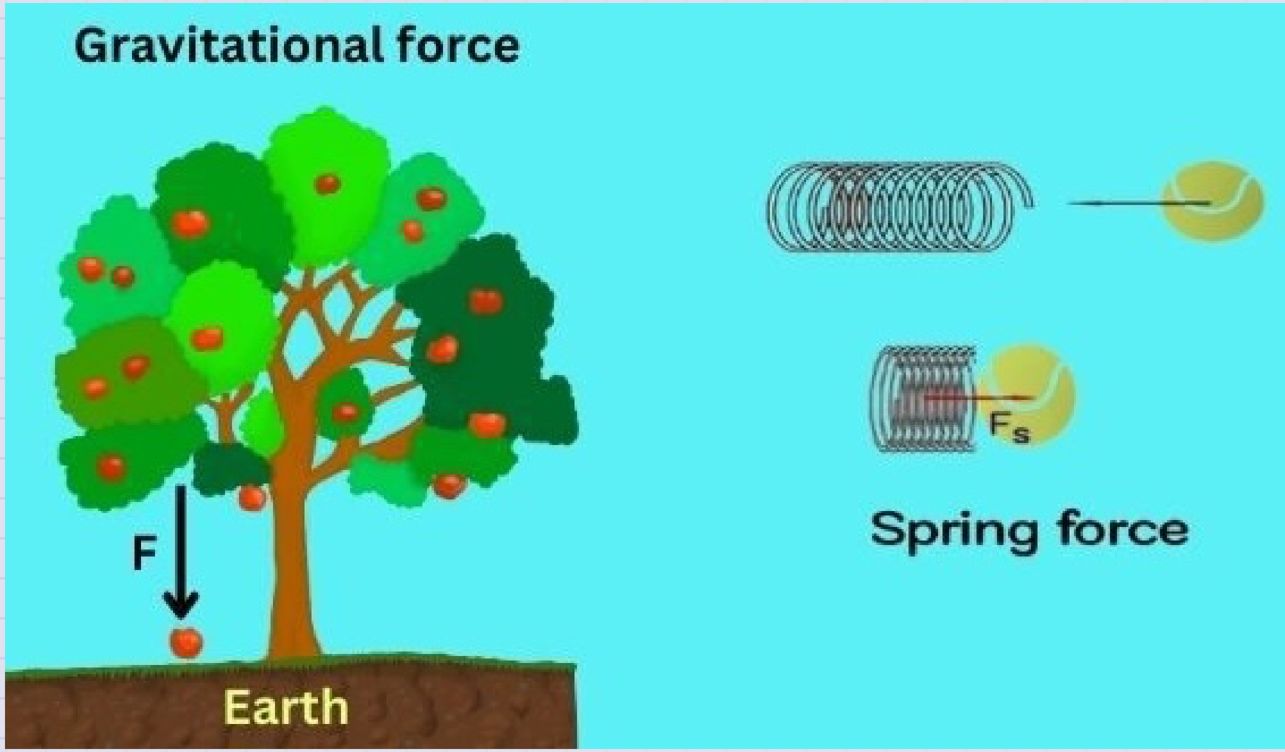
Day 13 - Conservative Forces

$$\vec{F} = -\nabla U$$

$$U = -\int \vec{F} \cdot d\vec{r}$$

$$\nabla \times \vec{F} = 0$$



Announcements

- HW 4 is due Friday
  - **Friday's Class:** We will work HW 4 Exercise 4 together + Q&A
  - Friday Office Hours (14:00-16:00; 1248 BPS)
- Midterm 1 is available now
  - We will talk about it Monday (come with questions)
  - There will be no office hours on Oct 3rd (DC traveling)

Seminars this week

WEDNESDAY, September 24, 2025

Astronomy Seminar, 1:30 pm, 1400 BPS, In Person and Zoom, Host~

Speaker: Kosuke Namekata, NAOJ

Title:

Zoom Link: https://msu.zoom.us/j/887295421?pwd=N1NFb0tVU29JL2FFSkk0cStpanR3UT09

Meeting ID: 887-295-421

Passcode: 002454

Seminars this week

THURSDAY, September 25, 2025 (Promotion Talk!)

Colloquium, 3:30 pm, 1415 BPS, in person and zoom. Host ~

Refreshments and social half-hour in BPS 1400 starting at 3 pm

Speaker: Wolfgang Kerzendorf, MSU - (PTRC)

Title: Calibrating Stellar Explosions as Probes of the Evolving Universe

Background:

For more information and to schedule time with the speaker, see the colloquium calendar at https://pa.msu.edu/news-events-seminars/colloquium-schedule.aspx

Zoom Link: https://msu.zoom.us/j/94951062663

Password: 2002 Or complete link: https://msu.zoom.us/j/94951062663?pwd=c48uM25P9UsRVuR74rkOioOWgpoxgC.1

Seminars this week

FRIDAY, September 26, 2025

QuIC Seminar, 12:30pm, -1:30pm, 1300 BPS, In Person

Speaker: Alexei Bazavov, MSU

Title: Efficient State Preparation for the Schwinger Model

Full Scheule is at: https://sites.google.com/msu.edu/quic-seminar/

For more information, reach out to Ryan LaRose

Seminars this week

FRIDAY, September 26, 2025

IReNA Online Seminar, 2:00 pm, In Person and Zoom, FRIB 2025 Nuclear Conference Room, Light refreshments will be served at 1:50pm.

Hosted by: Steffen Turkat (TU Dresden, Germany)

Speaker: Dominik Koll, HZDR, Germany

Title: The search for freshly synthesized radionuclides from stellar explosions on Earth

Zoom Link: https://msu.zoom.us/j/827950260

Password: JINA

This Week's Goals

- Remind ourselves of the concept of energy and energy conservation
- Apply the conservation of energy to a variety of systems
- Develop the mathematical tools to analyze energy conservation in more complex systems
- Connect our new understanding of energy conservation to our previous work on forces and motion

Reminders

- Energy is conserved in every process; our choice of system determines how we account for energy.
- Closed, isolated systems are often the simplest to analyze.
- A point particle is a model that allows us to ignore the internal structure of an object.
- The Work-Energy Theorem is just a statement of the conservation of energy for a point particle.

Conservation of Energy

General Principle: Energy is conserved in every process.

$$\Delta E_{sys} = W + Q$$

Isolated System: No work or heat is exchanged with the surroundings.

$$\Delta E_{sys} = 0$$

Point Particle: A model that allows us to ignore the internal structure of an object.

$$\Delta K = W_{\text{ext}}$$

The Potential Energy Function

Simple Harmonic Oscillator ($F_s = -kx$)

$$\Delta K = W_s$$

$$\frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2 = \int_{x_i}^{x_f} F_s dx = -\int_{x_i}^{x_f} kx dx$$

$$\frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2 = -\frac{1}{2}kx_f^2 + \frac{1}{2}kx_i^2$$

$$\frac{1}{2}mv_f^2 + \frac{1}{2}kx_f^2 = \frac{1}{2}mv_i^2 + \frac{1}{2}kx_i^2$$

$$K_f + U_{s,f} = K_i + U_{s,i}$$

$$U_s = \frac{1}{2}kx^2$$

Clicker Question 13-1

The gravitation force near the Earth's surface is given by $\vec{F} = -mg\hat{z}$. What is the potential energy function for this force? Choose $+\hat{z}$ to be up.

1. $U = -mgz$
2. $U = mgz$
3. $U = -mgz + U_0$
4. $U = mgz + U_0$
5. None of the above

Clicker Question 13-2

A model for a lattice chain acting on a electron is given by $F(x) = -F_0 \sin\left(\frac{2\pi x}{b}\right)$.

What is the potential energy function for this force?

1. $U = -F_0 \cos\left(\frac{2\pi x}{b}\right)$
2. $U = F_0 \cos\left(\frac{2\pi x}{b}\right)$
3. $U = -\frac{F_0 b}{2\pi} \cos\left(\frac{2\pi x}{b}\right)$
4. $U = \frac{F_0 b}{2\pi} \cos\left(\frac{2\pi x}{b}\right)$
5. None of the above

Clicker Question 13-3

I say "Stokes' Theorem" and you say...

1. HELL YEAH BROTHER 🤘
2. I'm not sure what that is 🤷
3. DEAR GOD WHY?!?! 😭

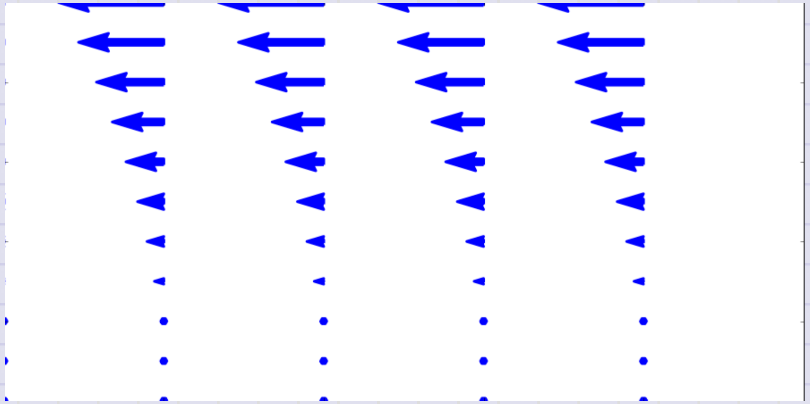
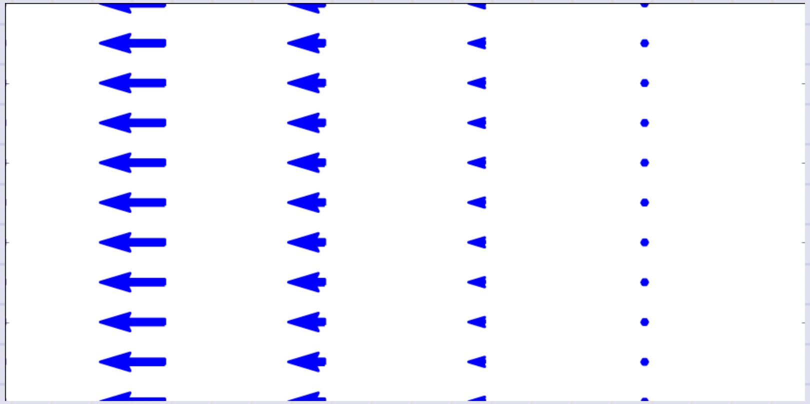
Clicker Question 13-4

The curl of a vector field is given by $\nabla \times \vec{F}$. If the curl of a vector field is zero, what can we say about the vector field?

1. It is a conservative force
2. It is a non-conservative force
3. It is a constant force
4. It is a force that does no work

Clicker Question 13-5

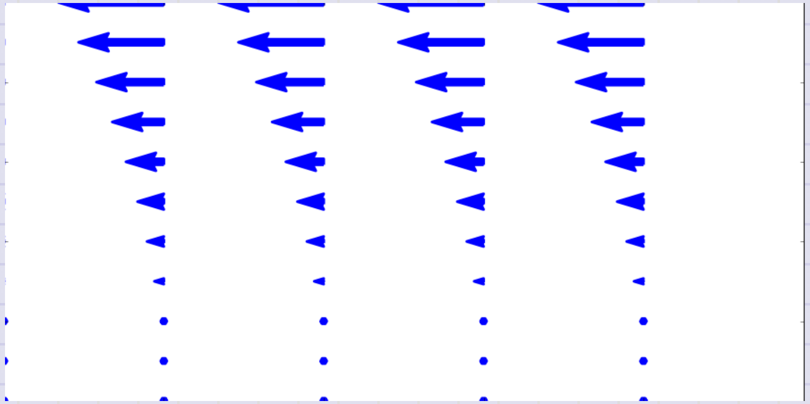
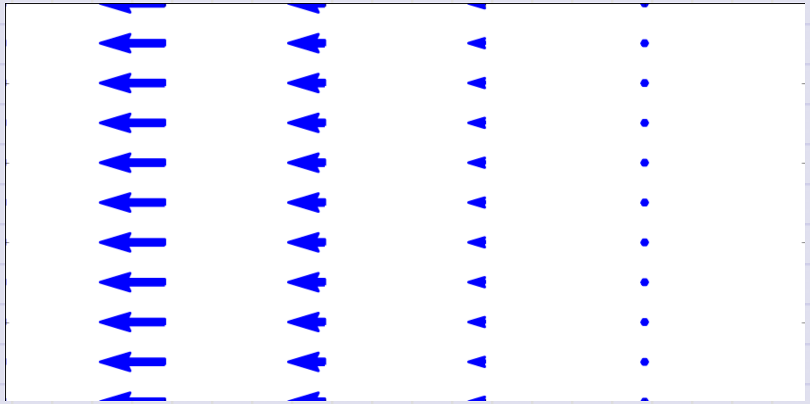
Which of the following fields have no divergence?

A.  B. 

1. A
2. B
3. Both A and B
4. Neither A nor B

Clicker Question 13-5

Which of the following fields have no curl?

A.  B. 

1. A
2. B
3. Both A and B
4. Neither A nor B

Clicker Question 13-6

Consider a vector field with zero curl: $\nabla \times \vec{F} = 0$. Which of the following statements is true?

1. The field is conservative
2. $\int \nabla \times \vec{F} \cdot d\vec{A} = 0$
3. $\oint \vec{F} \cdot d\vec{r} \neq 0$
4. $\vec{F}$ is the gradient of some scalar function, e.g., $\vec{F} = -\nabla U$
5. Some combination of the above